Fourier Transforms

DTFT and DFT

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Discrete Time Fourier Transform (DTFT)

- **The Fourier transform of a discrete time domain signal *x[n]*, also referred to as Discrete Time Fourier Transform (DTFT), is defined as**

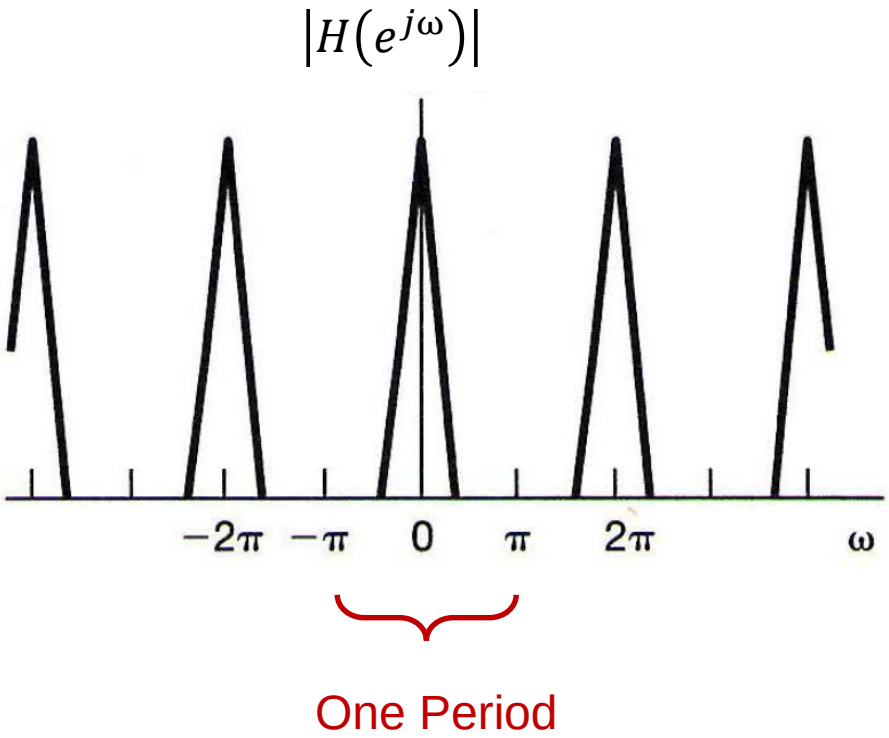
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- **We can recover *x[n]* back from *X(e^{jω})* using the inverse Fourier transform:**

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

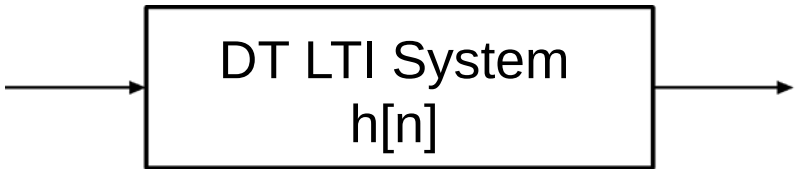
Frequency Response

- **The frequency response of a DT LTI system**
  - $H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$
    $= DTFT\{h[n]\}$
- **Continuous**
- **Periodic with period 2π**
- **Complex value**
  - $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta_H(e^{j\omega})}$
  - Magnitude: $|H(e^{j\omega})|$
  - Phase: $\theta_H(e^{j\omega})$



Frequency Response

$$x[n] = e^{j\omega_0 n} = cos(\omega_0 n) + jsin(\omega_0 n)$$



$$y[n] = h[n] * e^{j\omega_0 n} = \sum_k h[k] e^{j\omega_0 (n-k)}$$

$$= \sum_k h[k] e^{j\omega_0 n} e^{-j\omega_0 k} = e^{j\omega_0 n} \sum_k h[k] e^{-j\omega_0 k}$$

$$= e^{j\omega_0 n} H(e^{j\omega_0})$$

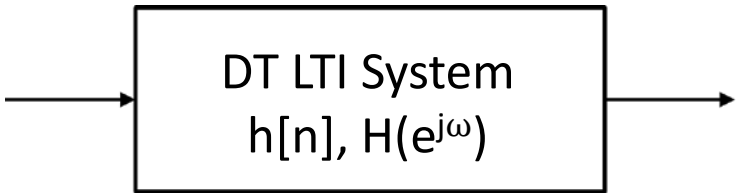
$e^{j\omega_0 n}$ is an eigen function, $H(e^{j\omega_0})$ is the eigenvalue.

$$y[n] = e^{j\omega_0 n} H\left(e^{j\omega_0}\right) = e^{j\omega_0 n} \left|H\left(e^{j\omega_0}\right)\right| e^{j\theta_H\left(e^{j\omega_0}\right)} = \left|H\left(e^{j\omega_0}\right)\right| e^{j\left(\omega_0 n + \theta_H\left(e^{j\omega_0}\right)\right)}$$

LTI system scales magnitude and changes phase, but NO new frequency!

Convolution Property of DTFT

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$$



$$y[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})o[e^{j\omega n}]d\omega =$$
$$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})H(e^{j\omega})e^{j\omega n}d\omega =$$
$$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})H(e^{j\omega})e^{j\omega n}d\omega =$$
$$\frac{1}{2\pi}\int_{-\pi}^{\pi} Y(e^{j\omega})\, e^{j\omega n}d\omega$$

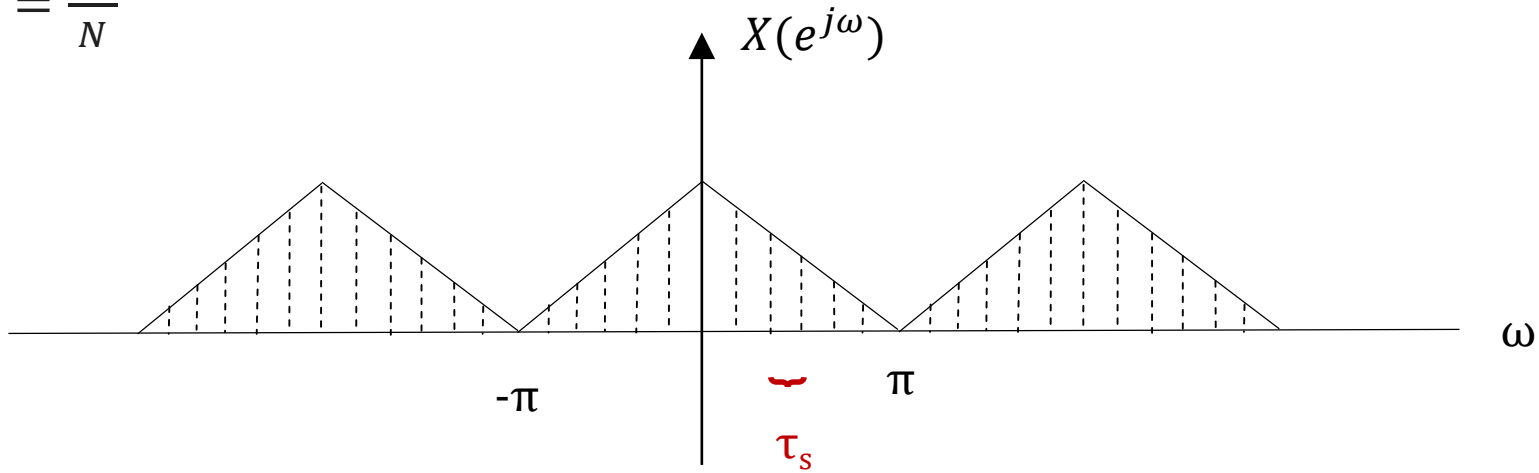
Convolution Property:

$$y[n] = x[n] * h[n] \overset{F}{\leftrightarrow} Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

Discrete Fourier Transform (DFT)

The DFT *X[k]* of a discrete time domain signal *x[n]* is a sampled version of the DTFT *X(e^{jω})*:

- $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$, continuous, periodic with period 2π
- τ_s: sampling period (spacing between adjacent samples in frequency domain)
- Assume *N* samples per period of 2π
- $\tau_s = \frac{2\pi}{N}$

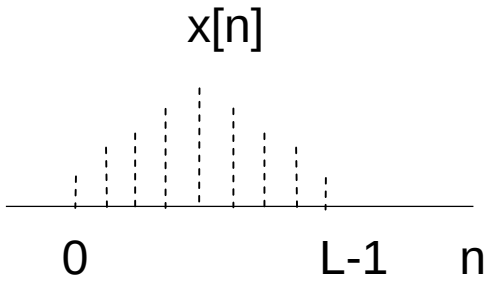


Discrete Fourier Transform (DFT)

- **Recall Nyquist Theorem: sampling in time domain gets periodicity in frequency domain.**
- **To avoid overlap, the time domain signal needs to be band-limited.**
- **Now sampling in frequency domain gets periodicity in time domain.**
- **To avoid overlap in *x[n]*, *x[n]* should have finite length.**

Discrete Fourier Transform (DFT)

- **Assume *x[n]* has finite length *L*.**

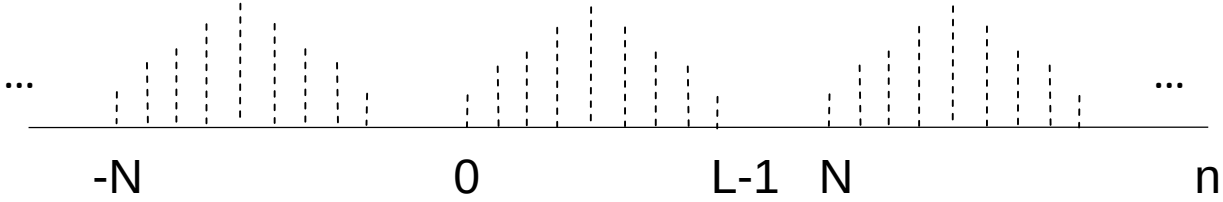


- **DTFT:** $X(e^{j\omega}) = \sum_{n=0}^{L-1} x[n]e^{-j\omega n}$

- **Sampling in one domain with a sampling period τ_s, replicates in the other domain with replication period $2\pi/\tau_s$.**

- **Replication of *x[n]* by period** $\frac{2\pi}{\tau_s} = \frac{2\pi}{\frac{2\pi}{N}} = N$**.**

- **To avoid overlap, the number of DFT samples *N* needs to be greater than or equal to the length *L* of signal *x[n]*:**
  - $N \geq L$



Discrete Fourier Transform (DFT)

- $X[k] = X\left(e^{j\omega}\right)\big|_{\omega_k = k\frac{2\pi}{N}} = \sum_{n=0}^{L-1} x[n]e^{-j\omega n}\big|_{\omega = k\frac{2\pi}{N}} = \sum_{n=0}^{L-1} x[n]e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$
- ***N*: size of DFT, *N* ≥ *L***
- **When *L* ≤ *N*, signal can be expanded by zero padding, so that**

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$$

Question

For a discrete-time signal of length L = 200, what is the minimum DFT size N to avoid overlap in the time domain given the four options below?

a) 128

b) 256

c) 512

d) 1024

Inverse Discrete Fourier Transform (IDFT)

- x[n] can be recovered using the inverse DFT:

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}, n = 0,1, \dots N-1$$

- Compare to DFT

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$$

- Form is similar, the algorithm used to compute DFT can be adjusted to compute IDFT.

DFT and Frequency Resolution

Digital frequency

- $\omega_k = k\frac{2\pi}{N}$ (in radians)
- Frequency resolution (spacing between frequency samples):
  - $\Delta\omega = \frac{2\pi}{N}$ (in radians)

Analog frequency

- $\Omega_k = \frac{\omega_k}{T_s} = \omega_k f_s = \frac{2\pi k}{N} f_s = \frac{2\pi k}{NT_s}$ (in radians per second)
- Frequency resolution (spacing between frequency samples):
  - $\Delta\Omega = \Delta\omega f_s = \frac{2\pi}{NT_s}$ (in radians per second)
  - $\Delta f = \frac{\Delta\Omega}{2\pi} = \Delta\omega\frac{f_s}{2\pi} = \frac{f_s}{N} = \frac{1}{NT_s}$ (in Hz)

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